# CO-1 → AT-3

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1. ## Smart Home automation System

(a) Types of intelligent agents:

1. Simple reflex agents

- Work on condition-action rules
- Respond only to the current percept (no memory).
- Example: motion sensor turning on a light when movement is detected.

2) model-based agent

- maintain an internal state (memory) of the environment
- use past information to make decision.
- Example: Temperature sensor agent that remembers previous temperature to predict heating/cooling needs.

3 Goal - based agent

- Act to acheive specific goals
- plan and choose actions to reach the goal.
- Example: AI agent that turns on the AC to reach desired temperature ~~to~~ or set by the user.

4. utility - based agent:

- Consider multiple possible outcomes and choose with highest utility.
- works with preference and trade-offs (eg, energy vs comfort).
- Example: Agent that adjusts temperature to balance comfort and energy efficiency based on utility value.

b) Justification of the most effective agent type for his system.

For a smart home automation system a utility - based agent is most effective. This is because its needs to balance multiple factor such as efficiency security and cost. unlike simply reflex or model - based agents can evaualte differentiely.

2. path to the exist in a maze (without prior knowledge).

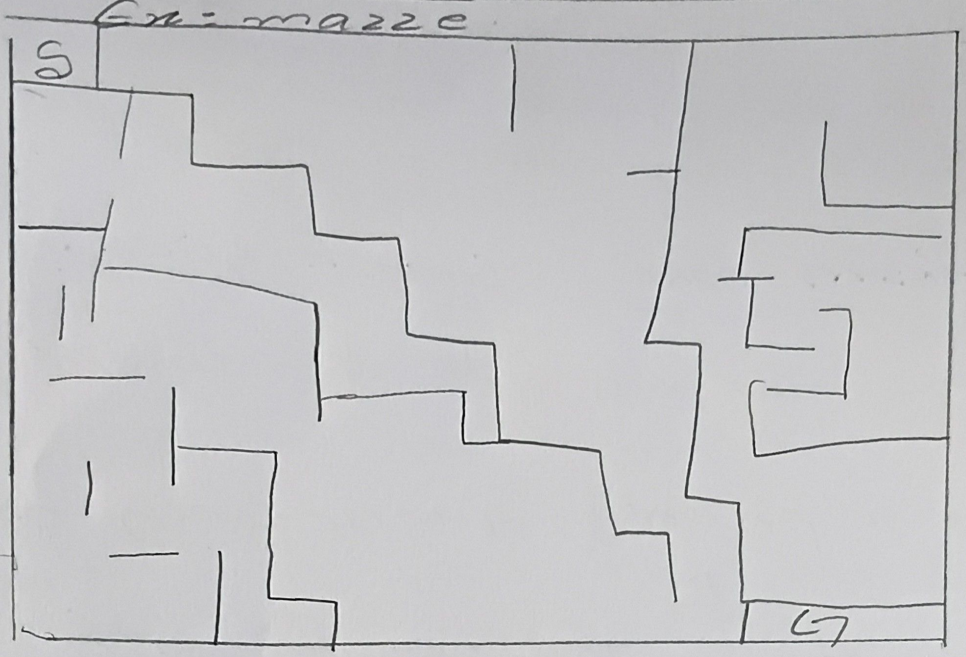
(a) uniform cost search (UCS)

- Explores the path with the lowest cost first.
- uses a priority queue (min-heap) based on path cost.
- Guarantees the optimal path (if all costs are positive)
- In a maze equal step cost; it behaves like BFS.

(b) Iterative deepening search (IDS)

- Combines the space efficiency of DFS and optimality of BFS.
- Increased depth limit step by step (0, 1, 2 ...).
- uses less memory than BFS.
- Good when the depth of the goal is unknown or large.

ex: mazze




Q) which combination is most effective why?

The best combination in UCS with IDS (or UCS + IDS) UCS ensure the optimal path based on cost, while IDS provides space coffeciency and works well when the goal depth is unknown together, they give a good balance of optimality, memory usage and reability in finding of cost.

Answer:

→ Most effective combination UCS + IDS (optimal path + space efficient).